

Forms of a complex number

Key Points

The cartesian form is

$$z = x + yi$$

The modulus, argument form is

$$z = |z|(\cos(\text{Arg}(z)) + i \sin(\text{Arg}(z)))$$

And using Euler's identity

$$e^{i\theta} = \cos \theta + i \sin \theta$$

We can write it in exponential form

$$z = |z|e^{i\text{Arg}(z)}$$

Basic Skills

Q1. Calculate the modulus and argument of the following complex numbers

- a) $6 + 8i$
- b) $-5 + 6i$
- c) $3 - i$

Q2. a) Find the modulus and argument of $z = 1 + i$

- b) Show that $z = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$
- c) By computing $\cos \frac{\pi}{4}, \sin \frac{\pi}{4}$ check your answer to b)
- d) Show that $z = \sqrt{2}e^{\frac{\pi i}{4}}$

Competency Skills

Q3. Complete the following table

Cartesian form	$5 - i$		
Modulus and Argument form		$2 \cos \frac{2\pi}{3} + 2i \sin \frac{2\pi}{3}$	
Exponential form			$\sqrt{3}e^{-\frac{3\pi i}{4}}$

Q4. Plot the complex number $z = \sqrt{3} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)$ on an argand diagram

Q5. Plot the complex number $z = 5e^{\frac{3\pi i}{7}}$ on an argand diagram

Advanced Skills

Q6. As an answer to a question a student writes $\cos \pi + i \cos -\frac{\pi}{2}$, while this is not in modulus and argument form it is still a complex number, plot it on an argand diagram

Q7. Given a number in exponential form $z = |z|e^{i\text{Arg}(z)}$ can you write its complex conjugate in exponential form

Q8. The exponential form of a complex number makes calculations easier, by first converting $z = 2(1 - i)$ and $w = 2i$ in exponential form compute the following

- a) z^2
- b) $z^2 w^3$
- c) $\frac{z}{2w}$
- d) $\frac{z}{2e^{0.5}}$

Extension

Q9. Explain using Euler's identity why $(\cos x + i \sin x)^n = \cos nx + i \sin nx$

Hint: Use the power in the identity in two different ways